

Data Sources & Provenance

OligoTox-Thrombocytopenia — every source, database and link behind the dataset

70 distinct sources · 1,959 measurements (1,453 human, 497 animal) · NIH/NCATS OligoTox Challenge, Phase 2

Every measurement in this dataset carries a `source_ref` and an exact `source_table` locus — the specific table, figure, claim or label section a value was read from. This document is **generated from the data**, so it cannot list a source the dataset no longer uses or omit one it does. For each entry: the full citation, the database it was retrieved from, a resolving link, the rights class governing reuse, the number of rows it contributed, and how many distinct loci within it were cited.

How to read the row counts. "Rows" is the number of per-measurement records drawn from that source; "loci" is the number of distinct places within it that were cited. A source with 387 rows across 268 loci was mined cell-by-cell, not summarised — the two numbers together show extraction depth.

Summary by database

Database	Retrieved via	Sources	Rows	Human
Peer-reviewed literature	europepmc.org · pmc.ncbi.nlm.nih.gov · doi.org	38	1,211	1,055
FDA — Drugs@FDA review documents	accessdata.fda.gov	12	271	201
EMA — EPAR assessment reports and SmPCs	ema.europa.eu	8	216	175
USPTO patents	patents.google.com	6	241	2
FDA — prescribing information (DailyMed)	dailymed.nlm.nih.gov	4	14	14
Trial registry	clinicaltrials.gov	2	6	6

Rights position across the dataset

Class	Meaning for reuse	Rows
Public domain	US Government work (FDA/EMA/USPTO) — values reproducible without restriction	748
CC-BY	Creative Commons Attribution — raw values reproducible with attribution	227
Summary stat.	Copyrighted; reproduced as summary statistics under fair use	984

Rights are tracked **per row**, not per dataset, so a consumer can filter to exactly the records they may lawfully reuse. A CC-BY classification is taken from the article's own licence field, never from the fact that it is free to read.

Peer-reviewed literature

Database: europepmc.org · pmc.ncbi.nlm.nih.gov · doi.org

Retrieval: Full text retrieved as JATS XML through NCBI E-utilities or the Europe PMC REST API, with supplementary files where present; publisher PDF or PubMed abstract where full text was not open.

Source	Link	Rows	Loci	Rights
PMID 28145801; PMCID PMC5467133; doi:10.1089/nat.2016.0650	—	387	387	Summary stat.
10.3324/haematol.2020.260059	—	157	32	Summary stat.
PMC5673186; doi:10.1371/journal.pone.0187574	—	115	20	CC-BY
PMC4322051; PMID 25646267; doi:10.1084/jem.20140391	—	91	38	Summary stat.
doi:10.3390/pharmaceutics15041112 (PMC10143489; PMID 37111598)	—	60	22	CC-BY
PMC8264460; PMID 33540294; doi:10.1016/j.thromres.2021.01.006	—	56	55	Summary stat.
10.1038/mt.2016.136 (PMC5112040; PMID 27357629)	—	49	47	Summary stat.
PMID 31390500; doi:10.1056/NEJMoa1715944	—	47	47	Summary stat.
PMC6386089 / PMID 30570431 / doi:10.1089/nat.2018.0753	—	43	43	Summary stat.
doi:10.1182/blood-2013-01-478966 (PMC3709655 / PMID 23673861)	—	37	22	Summary stat.
PMID:32243492; DOI:10.1093/cvr/cvaa077	—	22	20	Summary stat.
PMID:29972757; DOI:10.1056/NEJMoa1716793; PMCID:PMC12611561	—	19	19	Summary stat.
PMC11428464 / PMID 39335531 / doi:10.3390/biomedicines12092017 (Calcaterra IL et al., Biomedicines 2024;12:2017)	—	15	6	CC-BY
PMID 42381708 / PMCID PMC13315839 / doi:10.1016/j.omtn.2026.102976 — "Meta-analysis of adverse events in clinical studies with antisense oligonucleotide therapies", Mol Ther Nucleic Acids (CC BY)	—	11	10	CC-BY
PMC12846741 / PMID 41592789 / DOI 10.1093/nar/gkag007	—	11	5	CC-BY
PMC7835591 / PMID 33283485 / doi:10.1002/ehf2.13154 (Viney NJ et al., ESC Heart Fail 2021;8:652-661)	—	10	2	Summary stat.
PMID 28541820 / doi:10.1089/nat.2017.0666 (Henry et al. 2017, Nucleic Acid Ther 27:197-208)	—	6	4	Summary stat.
PMID 9152389 / doi:10.1016/s0022-3565(24)36667-4	—	6	1	Summary stat.
PMC10472096 / PMID 37364001 / doi:10.1158/1078-0432.CCR-22-2483 (Roschewski M et al., Clin Cancer Res 2023;29:3301-3312)	—	6	6	Summary stat.
PMC7063478 / PMID 31881528 / DOI 10.1016/j.omtn.2019.11.020	—	6	4	CC-BY
doi:10.1177/0192623314551840 — Frazier KS (2015) "Antisense oligonucleotide therapies: the promise and the challenges from a toxicologic pathologist's perspective", Toxicol Pathol 43:78-89	—	5	5	Summary stat.
PMID:9184201; doi:10.1016/s0300-483x(97)03661-5	—	5	2	Summary stat.
PMC12856967 / PMID 41085094 / doi:10.1002/cpdd.1615 (Youssef A et al., Clin Pharmacol Drug Dev 2026;15:879-907)	—	5	4	CC-BY
PMC9594513 / PMID 36270753 / doi:10.1136/bmjopen-2021-055718 (Nishina T et al., BMJ Open 2022;12:e055718)	—	5	5	Summary stat.
PMC9804925 / PMID 35971951 / doi:10.1002/cpdd.1154 (Han K et al., Clin Pharmacol Drug Dev 2022;11:1191-)	—	4	2	CC-BY
PMID 29846725 / doi:10.1093/toxsci/kfy119 (Narayanan et al. 2018, Toxicol Sci 164:613-626)	—	4	2	Summary stat.
PMID 37093125 / doi:10.1089/nat.2022.0042 (Shen et al. 2023, Nucleic Acid Ther 33:209-225)	—	4	2	Summary stat.

PMC6461157 / PMID 30817230 / doi:10.1089/nat.2018.0754 (Ferrone JD et al., Nucleic Acid Ther 2019;29:82-91)	—	4	4	Summary stat.
PMC8246029 / PMID 34222719 / doi:10.1016/j.jacbts.2021.04.004 (Morgan ES et al., JACC Basic Transl Sci 2021;6:485-)	—	4	4	Summary stat.
PMID 12269969 / doi:10.1046/j.1365-2036.2002.01341.x (Yacyshyn et al. 2002, Aliment Pharmacol Ther 16:1761-1770)	—	3	1	Summary stat.
PMID:22740102; DOI:10.1160/TH11-12-0889	—	3	3	Summary stat.
PMID:12438559; DOI:10.1124/jpet.102.036749	—	3	2	Summary stat.
PMID 39155024 / doi:10.1016/j.jtha.2024.08.003 (Johansson et al. 2024, J Thromb Haemost 22:3266-3276)	—	2	1	Summary stat.
PMID:28403626; DOI:10.1177/2048872617703065	—	2	1	Summary stat.
PMID:18025536; doi:10.1161/CIRCULATIONAHA.107.724864	—	1	1	Summary stat.
PMID 23836992 / PMC3699352 / doi:10.2147/OTT.S33077 (Higano CS. OncoTargets Ther. 2013;6:785-797)	—	1	1	Summary stat.
PMID 25782535 / PMC4574829 / doi:10.1111/bcp.12633 (Rabinovich-Guilatt L et al., Br J Clin Pharmacol 2015;80:436-45)	—	1	1	Summary stat.
PMID 30732140 / PMC6380863 / doi:10.1097/MD.00000000000014254 (Liu C, Li K. Medicine (Baltimore). 2019;98(6):e14254)	—	1	1	Summary stat.

FDA — Drugs@FDA review documents

Database: accessdata.fda.gov

Retrieval: Multi-discipline, clinical and pharmacology/toxicology reviews. Retrieved as PDF and parsed with PyMuPDF; a browser User-Agent is required or the host returns 404.

Source	Link	Rows	Loci	Rights
FDA NDA 211172 211172Orig1s000MedR.pdf	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=211172	115	109	Public domain
FDA NDA 217779 Orig1s000 Multi-discipline Review (imetelstat/Rytelo)	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=217779	91	88	Public domain
FDA NDA 203568 Orig1s000 PharmR (Kynamro/mipomersen Pharmacology Review(s))	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=203568	33	32	Public domain
FDA NDA 210922 Orig1s000 Multi-discipline Review (210922Orig1s000MultiR.pdf)	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=210922	8	7	Public domain
FDA NDA 217225 (IZERVAY / avacincaptad pegol) Pharmacology/Toxicology review, reviewer Maria I. Rivera PhD, Reference ID 5180757	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=217225	7	7	Public domain
FDA NDA 206488 Orig1s000 Pharmacology/Toxicology Review (EXONDYS 51, reviewer D.B. Hawver), Reference ID 3876740	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=206488	4	4	Public domain
FDA NDA 212194 Orig1s000 Multi-discipline Review and Evaluation (GIVLAARI), Reference ID 4522419	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=212194	3	3	Public domain
FDA NDA 214012 Orig1s000 Clinical Review (LEQVIO, reviewers E. Craig and I. Chowdhury), Reference ID 4697450	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=214012	3	3	Public domain
FDA NDA 213026 Orig1s000 Pharmacology/Toxicology Review (AMONDYS 45, reviewer B.J. Wilcox), Reference ID 4735654	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=213026	2	2	Public domain
FDA NDA 211970 Orig1s000 Pharmacology/Toxicology Review (VYONDYS 53, reviewer B.J. Wilcox), Reference ID 4457029	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&AppLN=211970	2	2	Public domain

FDA NDA 215515 Orig1s000 Clinical Review (AMVUTTRA, reviewer R.W. Paine MD PhD), Reference ID 4998267	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=215515	2	2	Public domain
FDA NDA 212154 Orig1s000 Pharmacology/Toxicology Review (VILTEPSO, reviewer D.L. Carbone), Reference ID 4641598	https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=212154	1	1	Public domain

EMA — EPAR assessment reports and SmPCs

Database: ema.europa.eu

Retrieval: European Public Assessment Reports and Summaries of Product Characteristics, retrieved as PDF and parsed with PyMuPDF.

Source	Link	Rows	Loci	Rights
EMA/180717/2019 (EMA/H/C/004538) Waylivra EPAR Public Assessment Report	https://www.ema.europa.eu/en/medicines/human/EPAR/waylivra	117	60	Public domain
EMA/411876/2018 (EMA/H/C/004782) Tegsedi EPAR public assessment report	https://www.ema.europa.eu/en/medicines/human/EPAR/tegsedi	81	58	Public domain
EMA_SmPC_WAYLIVRA_EPAR_product_information	https://www.ema.europa.eu/en/medicines	9	5	Public domain
EMA Oxlumio (lumasiran) EPAR public assessment report, EMA/568312/2020 (EMA/H/C/005040)	https://www.ema.europa.eu/en/medicines/human/EPAR/oxlumio	3	3	Public domain
EMA Qalsody (tofersen) EPAR public assessment report, EMA/276404/2024 (EMA/H/C/006072)	https://www.ema.europa.eu/en/medicines/human/EPAR/qalsody	3	2	Public domain
EMA Spinraza (nusinersen) EPAR public assessment report, EMA/289068/2017 (EMA/H/C/004312)	https://www.ema.europa.eu/en/medicines/human/EPAR/spinraza	1	1	Public domain
EMA Onpattro (patisiran) EPAR public assessment report, EMA/554262/2018 (EMA/H/C/004699)	https://www.ema.europa.eu/en/medicines/human/EPAR/onpattro	1	1	Public domain
EMA Amvuttra (vutrisiran) EPAR public assessment report, EMA/CHMP/689555/2022 (EMA/H/C/005852)	https://www.ema.europa.eu/en/medicines/human/EPAR/amvuttra	1	1	Public domain

USPTO patents

Database: patents.google.com

Retrieval: Worked-example tables and formal sequence listings. Retrieved as HTML from Google Patents; the USPTO print-PDF endpoint returns image-only scans with no text layer.

Source	Link	Rows	Loci	Rights
US 2023/0263822 A1	https://patents.google.com/patent/US20230263822A1/en	99	10	Public domain
US 11,535,849 B2	https://patents.google.com/patent/US11535849B2/en	68	6	Public domain
US 10,927,377 B2 (Ionis Pharmaceuticals, "Modulators of APOL1 expression", granted 2021-02-23)	https://patents.google.com/patent/US10927377B2/en	46	4	Public domain
US 2021/0355497 A1	https://patents.google.com/patent/US20210355497A1/en	10	4	Public domain
US 2015/0031747 A1	https://patents.google.com/patent/US20150031747A1/en	9	1	Public domain
US 10,100,310 B2 (Ionis Pharmaceuticals, "Modulation of prekallikrein (PKK) expression", granted 2018-10-16)	https://patents.google.com/patent/US10100310B2/en	9	1	Public domain

FDA — prescribing information (DailyMed)

Database: dailymed.nlm.nih.gov

Retrieval: Structured Product Labels retrieved as XML through the DailyMed SPL REST API and parsed directly.

Source	Link	Rows	Loci	Rights
FDA_label_TEGSEDI_SPL_8513207e-b55f-417b-9473-af785146a543_v10	https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=8513207e-b55f-417b-9473-af785146a543	6	3	Public domain
FDA_label_TRYNGOLZA_SPL_olezarsen	https://dailymed.nlm.nih.gov/dailymed/search.cfm?labeltype=all&query=TRYNGOLZA	4	2	Public domain
FDA_label_RYTELO_SPL_imetelstat	https://dailymed.nlm.nih.gov/dailymed/search.cfm?labeltype=all&query=RYTELO	2	2	Public domain
FDA_label_SPINRAZA_SPL_nusinersen	https://dailymed.nlm.nih.gov/dailymed/search.cfm?labeltype=all&query=SPINRAZA	2	1	Public domain

Trial registry

Database: clinicaltrials.gov
Retrieval: Posted results — structured adverse-event and outcome-measure tables, retrieved through the ClinicalTrials.gov API v2.

Source	Link	Rows	Loci	Rights
ClinicalTrials.gov NCT00138658 (posted results); primary publication PMID 22198426 / doi:10.1097/JTO.0b013e31823f459c (Laskin JJ et al., J Thorac Oncol 2012;7:579-86)	https://clinicaltrials.gov/study/NCT00138658	4	4	Public domain
ClinicalTrials.gov NCT00327340 (posted results); related publication PMID 21788353 / doi:10.1158/1078-0432.CCR-11-0859 (Saad F et al., CUOG trial P-06c, Clin Cancer Res 2011;17:5765-73)	https://clinicaltrials.gov/study/NCT00327340	2	2	Public domain

Retrieval routes, verbatim

Recorded so any value can be re-fetched, and because several routes are non-obvious — the wrong one silently returns nothing or an abstract stub:

Resource	Endpoint used
PMC open-access full text	<code>https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=pmc&id=<numeric id></code>
Europe PMC full text / supplementary files	<code>https://www.ebi.ac.uk/europepmc/webservices/rest/PMC<id>/fullTextXML – and /supplementaryFiles</code>
Europe PMC rendered PDF (<i>when XML is abstract-only</i>)	<code>https://europepmc.org/articles/PMC<id>?pdf=render</code>
PubMed abstract	<code>https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=pubmed&id=<pmid>&rettype=abstract</code>
DailyMed SPL (FDA labels)	<code>https://dailymed.nlm.nih.gov/dailymed/services/v2/spls.json?drug_name=<drug> → /spls/<setid>.xml</code>
FDA review documents	<code>https://www.accessdata.fda.gov/drugsatfda_docs/nda/<year>/<accession>.pdf (browser User-Agent required)</code>
EMA EPAR / SmPC	<code>https://www.ema.europa.eu/en/documents/assessment-report/<product>-epar-public-assessment-report_en.pdf</code>
Patents	<code>https://patents.google.com/patent/US<number>/en (USPTO print PDFs are image-only)</code>
Trial registry	<code>https://clinicaltrials.gov/api/v2/studies</code>
Citation resolution for this document	<code>https://www.ebi.ac.uk/europepmc/webservices/rest/search (resultType=core)</code>

What is deliberately absent

No third-party full text is redistributed. Sources are referenced by identifier and exact locus; the PDFs and XML retrieved during extraction were working files and are not committed. What is committed is the curation record — every agent's returned rows, the adversarial-verification verdicts, and the source sweep — so the published tables are reproducible from their inputs rather than merely re-checkable against these citations.

Sources consulted but not used are not listed here. Two categories were deliberately excluded during curation: on-target antithrombotic pharmacology (aptamers that inhibit platelet function *by design* — that is the intended mechanism, not toxicity), and papers whose platelet content could not be pinned to a specific locus. A row whose exact locus could not be named was dropped rather than kept with a vague citation.